Notes: Antràs and de Gortari (Econometrica, 2020)
On the Geography of Global Value Chains

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1 Big picture

- **Question.** How do trade barriers shape the geography of global value chains (GVCs), and which countries specialize in upstream versus downstream stages of production?
- **Core idea.** In a sequential value chain, the cost of moving goods is not the same at every stage. Trade costs compound as goods become more valuable, so downstream stages are more sensitive to trade costs than upstream stages.
- **Partial-equilibrium insight.** A firm choosing locations for N sequential stages cannot usually solve N separate location problems. With trade costs, the optimal location of one stage depends on where the good came from and where it will go next.
- **Computational insight.** The seemingly huge path problem can be decentralized and solved by dynamic programming. Instead of searching over J^N possible paths for each destination, one can solve a sequence of country-by-stage sourcing problems.
- **General-equilibrium insight.** In a Ricardian model with costly trade, countries' GVC positions are shaped jointly by productivity, wages/input costs, and geography. More central countries tend to specialize in more downstream stages, holding other determinants fixed.
- **Quantitative contribution.** The paper develops a tractable, estimable, multi-stage Eaton-Kortum-style model that can be mapped to World Input-Output Tables (WIOTs).
- **Estimation setting.** The quantitative exercise uses 2014 World Input-Output Database data for 43 countries plus the rest of the world. The model targets bilateral intermediate-input and final-good flows, domestic expenditure shares, country size, and value-added/gross-output patterns.
- **Main quantitative result.** Multi-stage production matters. The estimated model fits key input-output facts better than a roundabout one-stage model and produces larger gains from trade and larger losses from trade disruptions.
- **Economic takeaway.** The geography of GVCs is not just about who can produce each stage cheaply. It is also about where each country sits in the trading network and how trade costs accumulate along the chain.

2 Incremental contribution of the paper

- **From trade flows to production paths.** Standard quantitative trade models usually ask which country supplies a final good or an input. This paper studies the entire path of sequential production across countries.
- **A theory of GVC geography.** The paper gives formal content to the idea that location in the world trading network creates comparative advantage in different parts of the value chain.
- **Centrality-downstreamness nexus.** The paper shows that, other things equal, central countries tend to be more downstream. This prediction follows from the increasing importance of trade costs as a good moves toward completion.

- **A tractable multi-stage Ricardian model.** Previous multi-stage Ricardian models become hard to solve because stage-level Frechet draws do not aggregate neatly. The paper proposes alternative microfoundations that restore tractability and yield the same equilibrium equations.
- **A bridge to WIOT data.** The framework maps model objects to final-good flows, input flows, domestic expenditure shares, value-added shares, and gross-output shares in World Input-Output Tables.
- **A structural estimation contribution.** The paper combines calibration and GMM/MPEC-style estimation to back out trade costs, productivity terms, value-added shares, and sequential input shares from observed input-output data.
- **New counterfactual implications.** The paper shows that sequential production raises the measured cost of autarky and trade wars relative to a model without explicit GVC stages.
- **Policy relevance.** Because GVCs magnify the consequences of trade barriers, policy analysis that ignores sequential production can understate the income effects of protectionism and trade-war shocks.

3 Data, model objects, and measurement

3.1 Countries, stages, and value-chain paths

- There are J countries and N sequential stages of production.
- A production path is denoted by $\ell = \{\ell(1), \ell(2), \ldots, \ell(N)\}$, where $\ell(n)$ is the country performing stage n.
- Stage N is the most downstream stage and can be interpreted as final assembly.
- Consumers in each destination country buy final goods whose value chains may pass through many countries before completion.

Interpretation: The model's unit of analysis is not just a bilateral exporter-importer pair. It is an ordered sequence of production locations.

3.2 Sequential production technology

At each stage $n > 1$, production combines local composite inputs with the good completed up to stage $n-1$. A stylized version of the stage cost is:

$$p^n_{\ell(n)}(\ell) = \left(a^n_{\ell(n)} c_{\ell(n)}\right)^{\alpha_n} \left(p^{n-1}_{\ell(n-1)}(\ell)\tau_{\ell(n-1)\ell(n)}\right)^{1-\alpha_n}.$$

- a^n_i is the unit factor requirement for stage n in country i.
- c_i is the cost of the local composite factor in country i.
- τ_{ij} is the iceberg trade cost from i to j.
- α_n is the value-added or local-input share at stage n.

Interpretation: The good accumulates value as it moves down the chain. Therefore, moving the good later in the chain is more costly in value terms.

3.3 Compounding trade costs

Iterating the cost function gives a chain-level cost in which trade costs enter with stage-specific exponents:

$$p_j^F(\ell) \propto \prod_{n=1}^{N} \left(a^n_{\ell(n)} c_{\ell(n)} \right)^{\alpha_n \beta_n} \prod_{n=1}^{N-1} \tau^{\beta_n}_{\ell(n)\ell(n+1)} \tau_{\ell(N)j},$$

where

$$\beta_n = \prod_{m=n+1}^{N} (1 - \alpha_m).$$

Because β_n rises with n and $\beta_N = 1$, trade costs matter more in downstream stages.

Interpretation: A mistake in sourcing or shipping near the end of the chain is expensive because the good already embodies the value added from earlier stages.

3.4 World Input-Output Tables

- The empirical work uses World Input-Output Database data for 2014.
- The WIOD sample contains 43 countries plus a rest-of-world aggregate.
- A WIOT separates intermediate-input flows from final-good flows.
- The model is estimated to match entries of the WIOT, including bilateral input shipments, final-good shipments, gross output, and value added.

Interpretation: WIOTs are essential because GVCs are about intermediate production, not only final-good trade.

3.5 Country-level and trade-cost parameters

The quantitative model uses several key groups of parameters:

- country productivity parameters for intermediate and final goods;
- bilateral trade costs, calibrated from bilateral final-good flows using a Head-Ries-style approach;
- the Frechet shape parameter θ, chosen to match aggregate trade elasticities;
- country-level value-added shares γ_j;
- sequential input shares α_n^X and α_n^F for intermediate-input and final-good production.

Interpretation: The estimation separates ordinary country-level productivity and cost differences from the specifically sequential part of production.

4 Models and empirical specifications

4.1 Lead-firm cost minimization

A lead firm chooses the sequence of countries that minimizes the cost of serving destination j:

$$\ell_j = \arg\min_{\ell \in \mathcal{J}^N} p_j^F(\ell).$$

With no meaningful trade-cost geography, this would collapse to separate stage-by-stage choices. With arbitrary trade costs, the problem becomes a path problem: the best location for stage n depends on the locations of stages $n-1$ and $n+1$.

Interpretation: A country may not perform a stage even if it is the cheapest producer of that stage, because it may be far from the rest of the chain.

4.2 Dynamic programming and decentralization

The paper shows that the lead-firm problem can be decentralized. A producer at stage n in country i sources the stage-$(n-1)$ input from the least-cost source:

$$p_i^n = \min_{k \in \mathcal{J}} \left\{ (a_i^n c_i)^{\alpha_n} (p_k^{n-1} \tau_{ki})^{1-\alpha_n} \right\}.$$

- The decentralized equilibrium path coincides with the lead-firm cost-minimizing path.
- The computational burden falls from searching over all J^N paths to solving $J \times N \times J$ sourcing comparisons.
- This result relies on constant returns to scale.

Interpretation: The model becomes computationally feasible even with many countries and stages.

4.3 Ricardian general equilibrium

The paper embeds the sequential production problem in a multi-country Ricardian model inspired by Eaton and Kortum.

- Consumers have CES preferences over varieties.
- Labor is the primitive factor, and production also uses a bundle of materials.
- Country composite factor costs depend on wages and the local input price index.
- Productivity is represented so that the cost of a whole GVC path is Frechet-distributed.

Interpretation: This is a quantitative trade model with explicit sequential production and endogenous wages/input costs.

4.4 Path probabilities

The probability that a path ℓ is the cost-minimizing path for destination j is:

$$\pi_{\ell j} = \frac{\prod_{n=1}^{N-1} \left(\frac{T_{\ell(n)}}{c_{\ell(n)}}\right)^{\theta\alpha_n\beta_n} \tau_{\ell(n)\ell(n+1)}^{-\theta\beta_n} \left(\frac{T_{\ell(N)}}{c_{\ell(N)}}\right)^{\theta\alpha_N} \tau_{\ell(N)j}^{-\theta}}{\Theta_j}.$$

- $\pi_{\ell j}$ is analogous to the Eaton-Kortum trade share.
- When $N = 1$, the model collapses toward the standard one-stage Ricardian structure.
- When $N > 1$, trade costs along the whole path affect final sourcing probabilities.

Interpretation: A country's participation in a GVC depends not only on its own productivity and costs, but also on its location relative to other production stages and final demand.

4.5 Gains from trade

The welfare change from a change in trade costs can be expressed using the share of spending on purely domestic value chains:

$$\widehat{W}_j = (\widehat{\pi}_{jj})^{-1/(\theta\gamma)}.$$

Here π_{jj} is not simply the observed domestic expenditure share. It is the share of goods produced entirely through domestic supply chains.

Interpretation: The standard sufficient-statistic approach is not directly available because fully domestic value-chain shares are not observed in WIOTs. They must be inferred through the model.

4.6 Centrality and upstreamness

The paper defines an average upstreamness measure for countries and compares it to centrality. In the empirical implementation, centrality is constructed using proximity to other countries weighted by GDP or population.

- More upstream production occurs earlier in the chain.
- More downstream production occurs closer to final consumers.
- The model predicts that more central countries should be relatively downstream, all else equal.

Interpretation: Centrality is a source of comparative advantage because downstream stages are especially sensitive to being close to other production and consumption locations.

5 Identification, estimation, and counterfactual design

5.1 What the theory identifies

The theory identifies several mechanisms:

- trade costs compound along value chains;
- downstream stages are more trade-cost sensitive;
- country positions in GVCs depend on network geography;
- the economic gains from trade depend on whether production is one-stage or sequential.

Interpretation: The paper is not simply estimating a gravity equation. The gravity-style evidence is used as suggestive support; the main quantitative exercise is structural.

5.2 Suggestive empirical evidence

The paper presents two reduced-form empirical exercises before the structural estimation:

- final-good trade is more sensitive to distance than intermediate-input trade;
- more central countries tend to have lower export upstreamness, meaning they are relatively more downstream.

Interpretation: These facts line up with the model's key logic: downstream stages embody more accumulated value and therefore respond more strongly to trade costs.

5.3 Estimation strategy

The structural estimation proceeds in three broad steps:

- Use observed bilateral flows to infer bilateral trade costs.
- Fix the Frechet shape parameter θ to match trade elasticity evidence.
- Estimate the remaining model parameters by targeting the full WIOT structure.

The estimation compares versions with $N = 1, 2, 3, 4, 5$ stages and with asymmetric versus symmetric parameterizations.

Interpretation: The key comparison is whether allowing explicit sequential production improves the fit to input-output data and changes counterfactual welfare results.

5.4 Main estimated structure

- The asymmetric parameterization fits the data much better than the symmetric one.
- Sequential production appears especially important for intermediate inputs.

- The estimated model treats final-good production as effectively one-stage, while intermediate-input production benefits from multiple stages.
- Increasing from $N = 1$ to $N = 2$ generates a large improvement in fit; increasing beyond $N = 2$ adds much less.

Interpretation: The empirical action is in intermediate-input supply chains, which is exactly where GVC thinking is most relevant.

5.5 Counterfactuals

The paper conducts four main counterfactual exercises:

- **Autarky.** Shut down international trade and compute real income losses.
- **Zero gravity.** Remove trade barriers and compute potential free-trade gains.
- **U.S.-China trade war.** Increase U.S.-China bilateral trade barriers by 25 percent.
- **Regional versus global integration.** Scale trade costs up and down to see how domestic, NAFTA-regional, and global value chains change.

Interpretation: The counterfactuals show that sequential production changes both the magnitude and composition of the welfare consequences of trade-cost changes.

5.6 Limitations of the design

- The model is highly stylized and abstracts from many institutional details of actual GVCs.
- The baseline model assumes constant returns, which is central for the dynamic-programming solution.
- The model does not include explicit contracting frictions, firm boundaries, or multinational ownership choices.
- Some objects, especially fully domestic GVC shares, are inferred structurally rather than directly observed.
- The multi-industry extension is only preliminary because a full multi-industry, many-country, many-stage version is computationally demanding.

Interpretation: The paper's contribution is a tractable framework for thinking about sequential production, not a fully realistic model of every organizational feature of global supply chains.

6 Tables and figures

6.1 Figure 1 and Figure 2: Model logic and WIOT mapping

Read:

- Figure 1 reports simulations of the centrality-downstreamness relationship. The box plots show Spearman rank correlations between country centrality and average GVC position for different numbers of countries and stages.
- The correlations are mostly positive in the model's coding: more central countries tend to be more downstream.
- The relationship becomes stronger and tighter as the number of production stages increases.
- Figure 2 provides the schematic structure of a World Input-Output Table. The left block records intermediate-input flows and the right block records final-use flows.
- Rows represent sales by country-industry; columns represent input purchases by producing industries and final demand by consumers.

Economic message: Figure 1 gives the model's core prediction: geography matters more for downstream stages, so central countries become downstream specialists. Figure 2 shows why WIOT data are the right empirical object: they separately identify intermediate and final uses across countries.

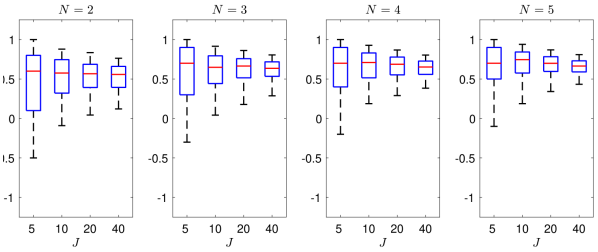


FIGURE 1.—Centrality downstreamness simulations.

(a) Figure 1: Centrality-downstreamness simulations

		Input use & value added			Final use			Total use
		Country 1	$\cdots$	Country J	Country 1	$\cdots$	Country J	
Output supplied	Country 1	$\boldsymbol{X}_{11}$	$\cdots$	$\boldsymbol{X}_{1J}$	$\boldsymbol{F}_{11}$	$\cdots$	$\boldsymbol{F}_{1J}$	$\mathbf{GO}_1$
	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$
	Country J	$\boldsymbol{X}_{J1}$	$\cdots$	$\boldsymbol{X}_{JJ}$	$\boldsymbol{F}_{J1}$	$\cdots$	$\boldsymbol{F}_{JJ}$	$\mathbf{GO}_J$
Value added		$\mathbf{GDP}_1$	$\cdots$	$\mathbf{GDP}_J$				
Gross output		$\mathbf{GO}_1$	$\cdots$	$\mathbf{GO}_J$				

FIGURE 2.—A schematic World Input–Output Table

(b) Figure 2: Schematic World Input-Output Table

6.2 Table I and Table II: Reduced-form evidence

Read:

- Table I estimates gravity-style trade-cost elasticities separately for total flows, final-good flows, and intermediate-input flows.
- The distance coefficient is more negative for final-good flows than for intermediate-input flows.
- Common language, contiguity, and domestic trade have positive effects, but these effects are generally attenuated for intermediate-input flows.
- Table II relates countries' export upstreamness to centrality and other controls.
- Centrality is negatively related to export upstreamness, which means more central countries tend to specialize in more downstream stages.

Economic message: The two tables provide empirical support for the model's key mechanism. Downstream trade is more sensitive to geography, and countries that are geographically central are more downstream in GVCs.

TABLE I

TRADE COST ELASTICITIES FOR FINAL GOODS AND INTERMEDIATE INPUTS[a]

	Total Flows		Final Good Flows		Input Flows	
	(1)	(2)	(3)	(4)	(5)	(6)
Distance	−1.111	−0.718	−1.229	−0.810	−1.058	−0.681
	(0.019)	(0.013)	(0.021)	(0.015)	(0.018)	(0.013)
Contiguity		1.149		1.161		1.154
		(0.090)		(0.098)		(0.088)
Language		0.402		0.499		0.358
		(0.024)		(0.028)		(0.023)
Domestic		5.320		5.610		5.061
		(0.168)		(0.191)		(0.159)
Observations	32,400	32,400	32,400	32,400	32,400	32,400
R^2	0.980	0.983	0.966	0.970	0.980	0.983

[a] Notes: Standard errors are clustered at the country-pair level. All regressions include exporter and importer fixed effects.

(a) Table I: Trade-cost elasticities

TABLE II

EXPORT UPSTREAMNESS AND CENTRALITY[a]

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Centrality (GDP weighted)			−0.173		−0.233		−0.155	
			(0.065)		(0.061)		(0.044)	
Centrality (pop. weighted)				−0.228		−0.239		−0.238
				(0.084)		(0.081)		(0.081)
$\log(Y/L)$	0.083	0.082	0.102	0.046	0.083	0.006		
	(0.142)	(0.142)	(0.138)	(0.148)	(0.046)	(0.037)		
Rule of Law	−0.029	−0.026	0.010	0.010				
	(0.103)	(0.104)	(0.105)	(0.105)				
Credit/Y	−0.437	−0.440	−0.375	−0.407				
	(0.136)	(0.137)	(0.130)	(0.135)				
$\log(K/L)$	0.156	0.159	0.163	0.188				
	(0.131)	(0.132)	(0.129)	(0.132)				
Schooling	−0.085	−0.085	−0.083	−0.094				
	(0.031)	(0.031)	(0.030)	(0.029)				
Observations	120	118	118	118	118	118	118	118
R^2	0.154	0.153	0.194	0.199	0.083	0.056	0.054	0.054

[a] Notes: Robust standard errors.

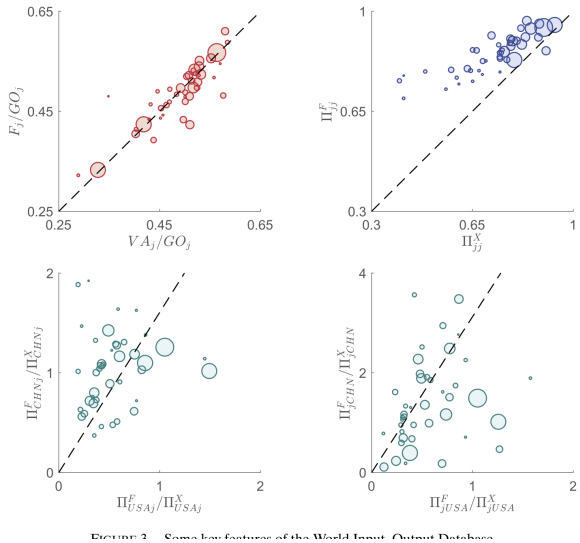
(b) Table II: Export upstreamness and centrality

6.3 Figure 3 and Table III: WIOD facts and estimation results

Read:

- Figure 3 plots key facts from the World Input-Output Database that the model is designed to match.
- The top panels show relationships between final output, value added, gross output, and domestic expenditure shares.
- The bottom panels show variation in relative final-good and intermediate-input import shares that one-stage roundabout models cannot fit well.
- Table III reports estimated production parameters for different numbers of stages and for asymmetric versus symmetric parameterizations.
- The objective function falls sharply when moving from a one-stage model to a two-stage sequential model, especially in the asymmetric case.
- The estimates imply that sequential production matters mainly for intermediate inputs, while final goods are effectively one-stage in the estimated asymmetric model.

Economic message: Figure 3 shows the data patterns that motivate the model, and Table III shows that the estimated sequential model fits these patterns much better than a roundabout benchmark.



(a) Figure 3: Key WIOD facts

TABLE III

ESTIMATION RESULTS

N	Obj. Func.	σ^X_{N-4}	σ^X_{N-3}	σ^X_{N-2}	σ^X_{N-1}	σ^X_N	σ^F_{N-4}	σ^F_{N-3}	σ^F_{N-2}	σ^F_{N-1}	σ^F_N	Mean γ_j
Asymmetric Parameterization												
1	39.44					1.00					1.00	0.51
2	12.07				1.00	0.25				1.00	1.00	0.62
3	11.68			1.00	0.03	0.17			1.00	1.00	1.00	0.71
4	11.57		1.00	0.02	0.04	0.14		1.00	1.00	1.00	1.00	0.77
5	11.50	1.00	0.02	0.04	0.05	0.10	1.00	1.00	1.00	1.00	1.00	0.82
Symmetric Parameterization												
1	55.28					1.00					1.00	0.55
2	37.88				1.00	0.21				1.00	0.21	0.88
3	37.88			1.00	1.00	0.21			1.00	1.00	0.21	0.88

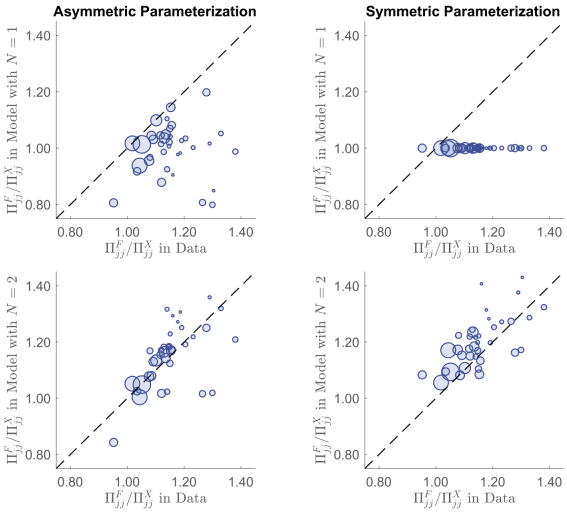
(b) Table III: Estimation results

6.4 Figure 4 and Figure 5: Model fit and autarky losses

Read:

- Figure 4 compares the ratio of domestic final-good expenditure shares to domestic input expenditure shares in the data and in the estimated model.
- The asymmetric one-stage model misses much of the cross-country variation; the asymmetric two-stage model fits substantially better.
- The symmetric parameterization performs worse because it restricts the model too tightly.
- Figure 5 reports welfare losses from reverting to autarky under estimated models with different numbers of stages.
- The losses are generally larger with sequential production than in the roundabout benchmark.
- Moving from $N = 1$ to $N = 2$ raises average autarky losses by roughly 60 percent.

Economic message: The model fit result shows that sequential production is empirically useful. The autarky result shows that GVCs magnify the welfare cost of shutting down trade because countries lose access to foreign comparative advantage at multiple stages.



(a) Figure 4: Model fit

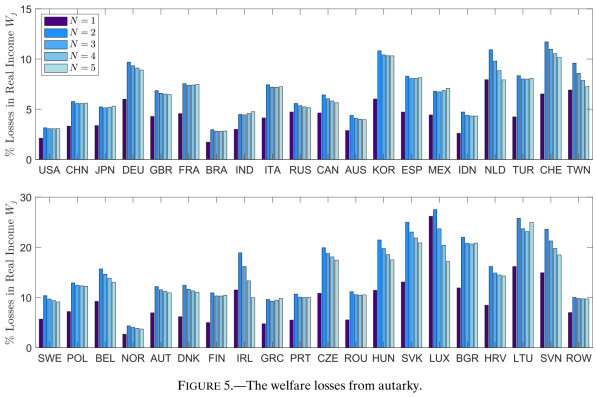


FIGURE 5.—The welfare losses from autarky.

(b) Figure 5: Welfare losses from autarky

6.5 Figure 6 and Figure 7: Free trade and the U.S.-China trade war

Read:

- Figure 6 reports welfare gains from completely eliminating trade barriers.
- The potential gains from free trade are much larger than the losses from autarky because removing all trade costs allows countries to exploit cross-country specialization more fully.
- Sequential production raises the gains for many countries. The paper reports GDP-weighted gains rising from about 225 percent with $N = 1$ to 276 percent with $N = 2$ and 317 percent with $N = 5$.
- Figure 7 studies a 25 percent increase in U.S.-China trade barriers.
- Both the United States and China lose, and the losses are larger in the sequential model than in the one-stage model.
- The model predicts substitution in U.S. sourcing: less from China, more from other Asian supply-chain economies and NAFTA partners, especially for intermediate inputs.

Economic message: Free trade and trade-war counterfactuals are more sensitive in a world of sequential production. Trade barriers do not just affect final goods; they disrupt where intermediate stages are performed and how supply chains reconfigure.

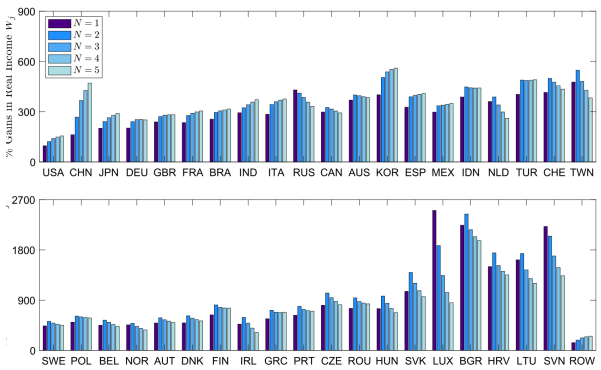


FIGURE 6.—The welfare gains from free trade.

(a) Figure 6: Welfare gains from free trade

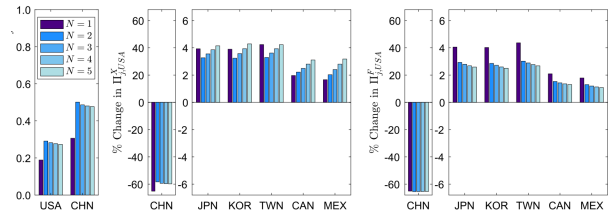


FIGURE 7.—The U.S.-China trade war.

(b) Figure 7: U.S.-China trade war

6.6 Figure 8 and Table IV: Regional integration and multi-industry extension

Read:

- Figure 8 studies the relative importance of domestic, regional, and global value chains for U.S. consumption as trade costs are scaled up or down.
- Domestic value chains are most important when trade costs are high; global value chains dominate when trade costs are low.
- Regional NAFTA value chains follow a hump shape: they rise when trade costs fall from high levels, but eventually decline as global chains become more attractive.
- Table IV presents a preliminary two-industry extension separating manufacturing and services.
- The table compares a roundabout model with several two-stage GVC specifications and reports welfare consequences of higher trade costs or autarky in manufacturing and services.
- The multi-industry exercise shows that manufacturing trade disruptions are generally more costly for moderate trade-cost increases, while services GVCs can generate very large autarky losses in some parameterizations.

Economic message: GVC integration can be regional at intermediate levels of trade costs but global at very low trade costs. Multi-industry structure matters because manufacturing and services enter supply chains differently.

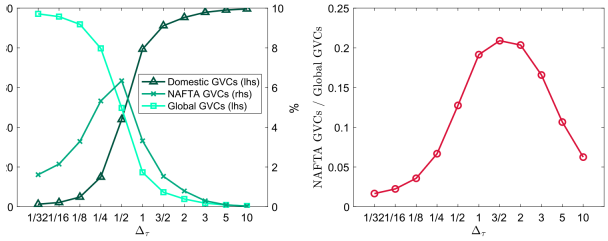


FIGURE 8.—Regional versus global integration.

(a) Figure 8: Regional versus global integration

TABLE IV

MULTI-INDUSTRY COUNTERFACTUALS

	Estimation Results					World Weighted Average $\hat{W}_j$					
	Obj. F.	$\alpha_2^{M,X}$	$\alpha_2^{S,X}$	$\alpha_2^{M,F}$	$\alpha_2^{S,F}$	$\hat{\tau}^M=2$	$\hat{\tau}^S=2$	$\hat{\tau}=2$	$\hat{\tau}^M=\infty$	$\hat{\tau}^S=\infty$	$\hat{\tau}=\infty$
Roundabout	49	1.00	1.00	1.00	1.00	3.7	1.0	4.6	4.9	1.1	6.0
GVC 1	34	0.00	0.18	1.00	1.00	3.6	1.6	5.1	4.8	1.9	6.7
GVC 2	40	1.00	0.40	1.00	1.00	3.7	2.4	5.9	4.9	3.0	7.7
GVC 3	37	0.00	0.93	1.00	1.00	3.6	3.0	6.5	4.8	23.1	26.8
GVC 4	39	1.00	0.91	1.00	1.00	3.8	3.4	7.0	5.0	21.7	25.5

(b) Table IV: Multi-industry counterfactuals

7 Short summary

- The paper develops a model of global value chains with multiple sequential stages and costly trade.
- Trade costs compound along the value chain because they apply to the gross value of goods being shipped.
- Downstream stages are therefore more sensitive to trade costs than upstream stages.
- A lead firm's path problem can be decentralized and solved with dynamic programming under constant returns to scale.
- In general equilibrium, geography becomes a source of comparative advantage: central countries tend to specialize in downstream stages.
- The paper builds a tractable Eaton-Kortum-style quantitative model with multi-stage production.
- The model is estimated using 2014 World Input-Output Database data for 43 countries plus rest of world.
- Sequential production is empirically important mainly for intermediate inputs.
- The sequential model fits key input-output facts better than a roundabout one-stage benchmark.
- Counterfactuals show larger gains from trade and larger losses from trade disruptions when GVC stages are explicitly modeled.

8 Conclusion

8.1 Core conclusion

Antràs and de Gortari show that the geography of GVCs is shaped by the interaction of productivity and trade costs along sequential production paths. The central result is that trade costs become more important as a good moves downstream. This gives central countries a comparative advantage in downstream stages and makes GVCs especially sensitive to trade barriers.

8.2 Economic intuition

The intuition is straightforward. Early in the chain, the good contains relatively little accumulated value. Later in the chain, the good embodies the inputs, labor, and productivity of earlier stages. Shipping the good later is therefore costly in a compounded way. Firms care especially about where downstream stages are located, and central locations become valuable because they economize on trade costs near the end of production.

8.3 Incremental contribution revisited

The paper advances the GVC literature in three ways. First, it provides a tractable theory of how geography shapes stage-level specialization. Second, it shows how to solve multi-stage sourcing

problems through decentralization and dynamic programming. Third, it brings the framework to WIOT data and shows that sequential production changes quantitative trade counterfactuals.

8.4 Implications for trade policy

Trade barriers have larger effects in a world of GVCs than in a one-stage world. A tariff or trade-cost increase can disrupt not only final-good trade but also the organization of intermediate stages. This means that trade wars can generate welfare losses through supply-chain reallocation, not just through final-demand channels.

8.5 Limitations

The framework is stylized. It abstracts from scale economies, contracting frictions, firm boundaries, multinational ownership, and the political economy of trade policy. It also requires structural inference for value-chain objects that are not directly observed. These limitations do not undermine the core contribution, but they indicate where future research can build.

8.6 Takeaway

The enduring takeaway is that GVCs are geographical paths, not just bilateral trade relationships. Once production is sequential, the same trade cost has different importance at different stages. This simple observation explains why central countries are downstream, why input-output data require explicit GVC structure, and why trade-cost shocks can have amplified welfare effects.